2 **(a)** Work done by a force is the product of the force and displacement in the direction of the force. [B1]

(b) <u>Consider a body of mass m that is raised to a vertical height h above the ground at a constant velocity by an external force F</u>. [B1]

Since velocity is constant, *net force = 0*

$$F - mg = 0$$

The magnitude of the external force, $F = mg$

<u>Work done by the external force, $W = F.s = mgh$</u> [B1]

Work done by F increases only the gravitational potential energy of the block

$$\text{Final } E_p - \text{Initial } E_p = mgh$$

If we take the gravitational potential energy at initial height to be the reference level (i.e. E_p at h_i = 0 J), then an object of mass m raised by height h vertically above the reference level has potential energy of

$\underline{E_p = mgh}$ (shown) [B1 for the entire highlighted portion]

(c) **(i)** By the conservation of energy,

loss in GPE of weight + loss in GPE of rod = work done against average resistance by ground

[C1, must clearly show that there are two losses in GPE. If calculation does not show this, zero marks for everything]

650(9.81)(1.2 + d) + 140 (9.81) (d) = 50 000 d [M1]

d = 0.18 m [A1]

(ii) During the collision, some <u>energy will be lost to the surroundings as thermal energy and sound energy</u> [B1], <u>resulting in less energy transferred</u> to the pole.

Can consider other specific context eg. Rocks or non-uniform soil

(For reference, the energy lost to surroundings can be due to inelastic collision, viscous forces doing work, the soil gaining kinetic energy, the rope not remaining slack and gaining elastic potential energy etc)